Algebra Monsoon 2026 Lecture 15 Cyclic Groups

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Cyclic groups

Def A group G is said to be **cyclic** if $G = \langle a \rangle$ for some $a \in G$. Such an a is called a **generator** of G.

- Every quotient of a cyclic group is cyclic.

If a group G is generated by an $a \in G$, the unique morphism of groups $f_a : \mathbb{Z} \to G$ such that $f_a(1) = a$ is **surjective**, and we have the isomorphism

$$\bar{f} : \mathbb{Z}/f_a^*(e) \xrightarrow{\sim} G.$$

As $f_a^*(e) = n\mathbb{Z}$ for a unique $n \geq 0$, we see that

- If G is infinite, then $n = 0$ and $f_a : \mathbb{Z} \to G$ is an isomorphism.
- If G is finite, then $n = o(G)$ and

$\overline{f_a} : \mathbb{Z}/n\mathbb{Z} \longrightarrow G$ is an isomorphism.

$\mathrm{Hom}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z})$.

Let us determine all morphisms of groups

$$f : \mathbb{Z}/m\mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z}$$

for integers $m \geq 0$, $n \geq 0$.

We have $m f([1]_m) = f([m]_m) = f([0]_m) = [0]_n$

We thus get a mapping

$$\mathrm{Hom}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) \longrightarrow m(\mathbb{Z}/n\mathbb{Z})$$

$$f \longmapsto \varphi \longrightarrow f([1]_m)$$

(where, for every module A and every integer $r \geq 0$, rA denotes the kernel of the endomorphism $a \mapsto ra$ of A).

As $\mathbb{Z}/m\mathbb{Z}$ is generated by $[1]_m$, the mapping φ is injective.

To show that φ is also surjective, let

$$x \in {}_m(\mathbb{Z}/n\mathbb{Z}).$$

There is a unique morphism

$$f : \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z}$$

such that $f(1) = x$. We have

$m\mathbb{Z} \subset \operatorname{Ker}(f)$, so there is a unique morphism

$$\bar{f} : \mathbb{Z}/m\mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z}$$

such that $\bar{f}([a]_m) = f(a)$ for every $a \in \mathbb{Z}$. In particular, $\bar{f}([1]_m) = x$, or equivalently $\varphi(\bar{f}) = x$, showing that φ is surjective. □

Question: What is the reciprocal bijection

$${}_m(\mathbb{Z}/n\mathbb{Z}) \longrightarrow \mathcal{G}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z})\ ?$$

$$\left(x \mapsto \left([a] \mapsto [ax]\right)\right)$$

$$[x]_n \longmapsto \left([1]_m \Rightarrow [x]_n \right).$$

In particular, for $m = n$, we have a bijection

$$\mathrm{End}_{\mathcal{G}}(\mathbb{Z}/n\mathbb{Z}) \xrightarrow{\sim} \mathbb{Z}/n\mathbb{Z}$$

This bijection is a morphism of multiplicative monoids:

if $f([1]_n) = a$ and $g([1]_n) = b$,

then $(g \circ f)([1]_n) = g(a)$

$= ab$.

Therefore, the mapping $f \longmapsto f([1]_n)$ is an isomorphism of multiplicative monoids.

In particular, we get an isomorphism

$$\mathrm{Aut}_{\mathcal{G}}(\mathbb{Z}/n\mathbb{Z}) \xrightarrow{\sim} (\mathbb{Z}/n\mathbb{Z})^{\times}$$

□

of groups.

Exercise: Show that $\mathrm{End}_g(\mathbb{Z}) \longrightarrow \mathbb{Z}$
$f \longmapsto f(1)$
is an isomorphism of multiplicative monoids.
In particular, $\mathrm{Aut}_g(\mathbb{Z}) = \mathbb{Z}^\times$. □

Subgroups of $\mathbb{Z}/m\mathbb{Z}$

We have seen that

(subgroups of $\mathbb{Z}/m\mathbb{Z}$)
$\xrightarrow{\sim}$ (subgroups of $\mathbb{Z}$ containing $m\mathbb{Z}$)
$H/m\mathbb{Z} \longleftarrow\!\shortmid H$

is a bijection. We have also seen that subgroups of $\mathbb{Z}$ are of the form $d\mathbb{Z}$ for a unique $d \geq 0$. We have

$m\mathbb{Z} \subset d\mathbb{Z} \iff m$ is a multiple of d.

$m\mathbb{Z} \subset d\mathbb{Z} \iff$ d is a divisor of m.

As the order of $d\mathbb{Z}/m\mathbb{Z}$ is m/d, we get a subgroup of order m/d (index d) in $\mathbb{Z}/m\mathbb{Z}$.

Therefore

- Let C be a cyclic group of order $m \neq 1$.
  For every divisor d of m, there is a unique subgroup of order d, namely ^dC; it is <u>cyclic</u>:
  if $C = \langle x \rangle$, then $^dC = (x^{m/d})$.

Let $m \neq 1$ be an integer and $a \in \mathbb{Z}$.

<u>Proposition</u> $\langle a \rangle = \mathbb{Z}/m\mathbb{Z}$

$\iff \gcd(a, m) = 1$.

<u>Proof</u> Put $d = \gcd(a, m)$. If $\langle a \rangle = \mathbb{Z}/m\mathbb{Z}$,
there exists a $u \in \mathbb{Z}$ such that

$au \equiv 1 \pmod{m}$, so

$au + mv = 1$ for some $v \in \mathbb{Z}$,

and therefore $\gcd(a, m) = 1$.

Conversely, if $\gcd(a, m) = 1$, then

exist $u, v \in \mathbb{Z}$ such that

we then have $au \equiv 1 \pmod{n}$, and

therefore $\langle a \rangle = \mathbb{Z}/n\mathbb{Z}$.

Proposition Let A, B be cyclic groups of orders $m \geq 1$, $n \geq 1$.

If $\gcd(m, n) = 1$, then $A \times B$ is cyclic.

Proof We have $o(A \times B) = mn$.

Write $A = \langle a \rangle$ and $B = \langle b \rangle$ and consider $c = (a, b)$ in $A \times B$.

For every $k \in \mathbb{Z}$, we have

$$c^k = (a, b)^k = (a^k, b^k).$$

So $c^k = e \iff a^k = e$ and $b^k = e$,

hence k is a multiple of $\operatorname{lcm}(m, n)$.

But $\gcd(m, n) = 1$, so $\operatorname{lcm}(m, n) = mn$.

In other words c has order mn,

so $A \times B = \langle c \rangle$. □

and hence $\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$.

Conversely, if A, B are cyclic of orders m, n and if $\gcd(m,n) > 1$, then $A \times B$ is not cyclic.

Indeed, if p is a prime dividing $\gcd(m,n)$, then there are subgroups $C \subset A$, $D \subset B$ of order p, and $C \times D$ is a subgroup of $A \times B$. As $C \times D$ is not cyclic (every element is killed by p, so there is no element of order $p^2 = o(C \times D)$), nor is $A \times B$ (subgroups of cyclic groups are cyclic).